

SHUAIHANG CHEN

PhD Student in Robotics

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SUMMARY

PhD student in robotics working on vision-based robotic manipulation and Vision-Language-Action (VLA) post-training. My recent work focuses on improving the real-world robustness and data efficiency of embodied foundation models using reinforcement learning, minimal real-world data, and large-scale simulation under substantial sim-real mismatch.

RESEARCH EXPERIENCE

Sim-Real Co-Training for Vision-Based Robotic Manipulation

2025

[Project Page](#) [Paper](#)

- Drove end-to-end OpenVLA tuning across four real-world manipulation tasks, improving average success rate from 40% to 64% (+24 points) over the behavior-cloning-based Sim-and-Real Co-Training baseline.
- Developed a two-stage post-training recipe: supervised fine-tuning on mixed real/sim demonstrations, followed by reinforcement learning in simulation with real-world regularization.
- Built the simulation environments for two of the four tasks, supporting RL-Co post-training, evaluation, and sim-to-real transfer under large sim-real mismatch.
- Designed data-scaling ablations showing that, under the evaluated setting, RL-Co with 50 real trajectories outperformed pure SFT trained on 200 real trajectories.
- Showed that effective post-training did not require realistic simulation: real trajectories only needed to be replayable in simulation, despite substantial gaps in objects, scenes, lighting, textures, and physical parameters.

Value Modeling for $\pi_{0.6}$ -Style VLA Post-Training

2026

[RLinf](#) [GitHub](#)

- Reproduced the $\pi_{0.6}$ RECAP training and evaluation pipeline on RLinf for VLA post-training.
- Compared multiple value-model designs and identified the most stable configuration under the current setting.
- Diagnosed large variance in CFG-based gains, which could fluctuate sharply or become negative, indicating that current improvements still rely heavily on data scale and hyperparameter tuning.

EDUCATION

Zhongguancun Academy & Harbin Institute of Technology

2024 – Present

Ph.D. Student in Robotics (Joint Program)

Advisor: [Chao Yu](#)

Harbin Institute of Technology

2020 – 2024

B.Eng. in Software Engineering

EXPERTISE

Research VLA Post-Training, Vision-Based Robotic Manipulation, Sim-Real Co-Training under Large Sim-Real Gaps, Value Modeling

Systems ManiSkill simulation and task setup, RLinf-based RL training pipelines, OpenVLA tuning and evaluation, real-to-sim trajectory replay